

Introduction to Git

Git Repository Basics

Learning objective

Understand Git, version control, repositories, and why developers use them.

Core idea

Git is a distributed version control system that records how project files change over time. It helps developers review work, restore earlier versions, create separate branches, and collaborate without overwriting each other's changes.

A **Git repository** is a project folder tracked by Git. It contains the project files and a hidden **.git** folder that stores history and configuration.

Git vs. GitHub: Git is the version control tool installed on your computer. GitHub is an online platform for storing and sharing Git repositories.

Commands to know

Command / pattern	Purpose
<code>git --version</code>	Check whether Git is installed and view its version

Step-by-step workflow

- 1. Open Terminal, Command Prompt, or Git Bash.
- 2. Run **git --version**.
- 3. Confirm that a Git version is displayed.
- 4. Continue to Git configuration and repository initialization.

Common mistakes

- Assuming Git and GitHub are the same product.
- Expecting Git to upload files automatically.
- Editing or deleting the hidden **.git** folder manually.

Practice task

Open your terminal and run **git --version**. Example output: `git version 2.50.1.windows.1`. Your installed version may be different.

Quick check

What is the difference between Git and GitHub?

Tip: Run `git status` before and after important Git commands so you can observe what changed.